

Topic -2 (Part 1): Multi-armed Bandits

AIMLCZG512-Deep Reinforcement Learning

Lecture Notes

1. Topics Covered

1. The k -armed bandit formulation as a basic evaluative-feedback problem.
2. True action values, estimated action values, and sample-average estimation.
3. Greedy action selection and the limitation of pure exploitation.
4. ϵ -greedy action selection as a simple exploration mechanism.
5. Modelling repeated one-step decision problems using the bandit framework.
6. Experimental interpretation of the 10-armed testbed.
7. Key bandit terminology, numerical practice, and required reading.

2. A k -armed Bandit Problem

A k -armed bandit problem is a simplified **evaluative-feedback** problem with one decision situation and k available actions. At each time step, the learner chooses one action and receives a numerical reward generated by the reward distribution associated with that selected action. The objective is to maximize the expected total reward over many action selections.

Formal setup. At time step t , the learner selects an action A_t from the set of k available actions and receives a reward R_t . Each action a has a true action value

$$q_*(a) \doteq \mathbb{E}[R_t \mid A_t = a],$$

which is the expected reward obtained when action a is selected.

The true action values $q_*(a)$ are not known to the learner. The learner therefore maintains estimates, denoted by $Q_t(a)$, and improves these estimates from the rewards observed after selecting actions.

Basic characteristics. In the basic bandit setting, there is a single situation, a fixed set of k actions, reward feedback only for the selected action, unknown true action values, and a direct conflict between selecting the currently best-looking action and trying other actions to improve the estimates.

The bandit setting is important because it isolates one essential difficulty of reinforcement learning: the feedback is **evaluative**, not instructive. The reward evaluates the action actually taken, but it does not state whether that action was the best possible action.

3. Action-Value Methods

An **action-value method** estimates the value of each action and then uses those estimates to select actions. In this introductory setting, the most direct estimate is the average reward received after selecting that action.

$$Q_t(a) = \frac{\text{sum of rewards received after selecting } a \text{ before } t}{\text{number of times } a \text{ was selected before } t}.$$

This is called a **sample-average estimate**. If action a has been selected many times, its estimate is based on more evidence. If it has been selected only once or twice, its estimate may still be unreliable.

3.1. Repeated Estimation Cycle

The same four-step loop is repeated at every action selection:

1. **Maintain estimates:** keep a current estimate $Q_t(a)$ for each action.
2. **Select an action:** choose one action using an action-selection rule.
3. **Observe reward:** receive the numerical reward for the selected action.
4. **Update only that action:** revise the estimate of the selected action using the newly observed reward.

The loop then repeats at the next time step. The estimates are learned from experience, but each update uses information only about the action actually selected, not about the unselected actions.

3.2. Numerical Illustration of Sample-Average Estimation

Assume three actions have produced the following observed rewards:

Action	Observed rewards	Sample-average estimate
a_1	2, 4, 3	$Q(a_1) = \frac{2+4+3}{3} = 3.00$
a_2	5, 1	$Q(a_2) = \frac{5+1}{2} = 3.00$
a_3	0, 2, 1, 1	$Q(a_3) = \frac{0+2+1+1}{4} = 1.00$

Here a_1 and a_2 are tied under the present estimates. Further observations may be needed to decide which action is actually better.

4. Greedy Action Selection

A **greedy action** is an action whose current estimated value is highest. Greedy action selection chooses such an action:

$$A_t \doteq \arg \max_a Q_t(a).$$

This rule is simple and uses current knowledge fully. Its weakness is that an early lucky or unlucky reward can mislead the learner. If the learner stops trying alternatives too early, it may continue selecting a suboptimal action.

Example. Suppose $Q_t(a_1) = 1.8$, $Q_t(a_2) = 2.4$, and $Q_t(a_3) = 1.9$. A greedy method selects a_2 because it has the highest current estimate.

Greedy action selection is useful as a baseline method. However, by itself it does not provide a mechanism for deliberately testing less-known actions.

5. ϵ -Greedy Action Selection

An ϵ -greedy method behaves greedily most of the time, but occasionally explores. With probability $1 - \epsilon$, it selects a greedy action. With probability ϵ , it selects randomly from all k actions.

Rule. With probability $1 - \epsilon$, select an action with the highest current estimate. With probability ϵ , select one of the k actions at random.

This rule gives a simple practical balance. The learner mostly uses its current estimates, but it still gives every action some chance of being selected.

5.1. Numerical Illustration of ϵ -Greedy Selection

Assume a 4-armed bandit with current estimates:

$$Q_t(a_1) = 1.5, \quad Q_t(a_2) = 2.1, \quad Q_t(a_3) = 1.2, \quad Q_t(a_4) = 0.9.$$

The greedy action is a_2 . Let $\epsilon = 0.20$.

Quantity	Value and meaning
Probability of random exploration	$\epsilon = 0.20$. A random action is selected 20% of the time.
Probability of direct greedy choice	$1 - \epsilon = 0.80$. The method directly selects a_2 80% of the time.
Probability of each action during random selection	$\epsilon/k = 0.20/4 = 0.05$. Each action receives 5% probability through the random part.
Total probability of selecting a_2	$(1 - \epsilon) + \epsilon/k = 0.80 + 0.05 = 0.85$.
Total probability of each non-greedy action	$\epsilon/k = 0.05$. Thus a_1 , a_3 , and a_4 each receive probability 0.05.

Interpretation. The greedy action can also be selected during the random exploration step. Therefore, when there is a single greedy action, its total probability is $1 - \epsilon + \epsilon/k$.

6. Bandit Problem Formulations

A problem can be treated as a bandit problem when the same learner repeatedly chooses one option from a fixed set and receives scalar reward feedback for only the chosen option.

Problem setting	Actions or arms	Reward signal and bandit interpretation
Online advertisement selection	Candidate advertisements to show to a visitor.	Reward may be click, purchase, or revenue. The learner estimates which advertisement gives the best average return.
Website layout testing	Alternative page layouts, button positions, or message variants.	Reward may be conversion, sign-up, or successful task completion. Each layout is treated as one arm.
Treatment choice in a controlled setting	Candidate treatment options for comparable cases.	Reward may be improvement score, recovery indicator, or reduction in symptoms. The selected treatment alone gives feedback.
Cloud configuration tuning	Alternative server, cache, or resource-allocation configurations.	Reward may be throughput, negative latency, or cost-adjusted performance. The system learns which configuration performs best on average.

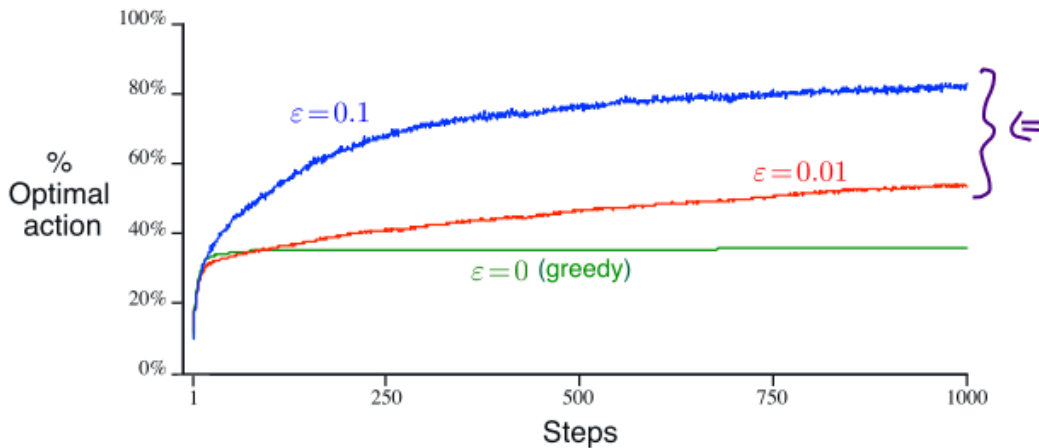
In all these examples, the learner does not observe the reward that would have been obtained from the unselected alternatives. This missing information is what makes exploration necessary.

7. The 10-armed Testbed

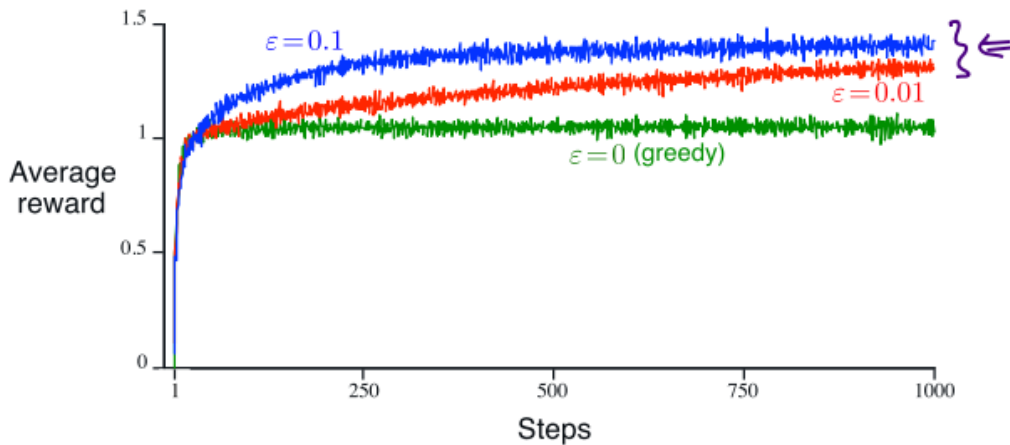
The 10-armed testbed is an experimental setting used to compare action-value methods. Its purpose is to show how different action-selection rules behave when rewards are uncertain and true action values are not directly known.

In the testbed, each bandit problem has $k = 10$ actions. The true value $q_*(a)$ of each action is chosen from a normal distribution with mean 0 and variance 1. When the learner selects action A_t , the reward R_t is sampled from a normal distribution whose mean is $q_*(A_t)$ and whose variance is 1. Thus the reward is noisy even when the true action value is fixed.

The reported curves are averages over many independently generated bandit problems. Each method is run for 1000 steps, and the results summarize how its behaviour improves with repeated action selections.



Percentage of times the optimal action is selected over 1000 steps.



Average reward over 1000 steps for greedy, $\epsilon = 0.01$, and $\epsilon = 0.1$ methods.

The optimal-action curve shows that the greedy method remains low because it does not continue testing alternatives. The $\epsilon = 0.1$ method discovers the optimal action faster because it explores more often. The $\epsilon = 0.01$ method improves more slowly because it explores less frequently.

The average-reward curve shows a similar pattern. The greedy method improves quickly in the beginning but settles at a lower reward. The ϵ -greedy methods perform better over time because occasional exploration helps correct early misleading estimates.

7.1. Lessons from the Testbed

- **Early estimates can mislead:** a good action may look poor if its first observed reward is low.
- **Greedy selection can settle too early:** once it commits, it may not return to actions that were initially underestimated.
- **ϵ -greedy selection improves reliability:** occasional random selection gives less-tested actions a chance to be evaluated.
- **Exploration has a cost:** random actions may reduce immediate reward, but they can improve long-run reward.

8. Review of Key Terms

Term	Short definition
k -armed bandit	A repeated one-situation decision problem with k available actions and reward feedback after each selected action.
Arm	One selectable option in a bandit problem.
Evaluative feedback	Feedback that evaluates the selected action but does not identify the best possible action.
Action-value method	A method that estimates the values of actions and uses those estimates for action selection.
True action value, $q_*(a)$	The expected reward obtained when action a is selected.
Estimated action value, $Q_t(a)$	The learner's current estimate of the true value of action a at time t .
Sample-average estimate	An estimate formed by averaging rewards previously observed after selecting the same action.
Action-selection rule	The rule used to choose an action from the current estimates.
Greedy action	An action with the highest current estimated value.
ϵ -greedy selection	An action-selection rule that chooses greedily most of the time and randomly with probability ϵ .
Optimal action	The action with the highest true expected reward.
10-armed testbed	A 10-action experimental setting used to compare bandit action-selection methods.

9. Review Questions

1. Define the k -armed bandit problem. State the learner's objective and explain why the problem is described as a one-situation decision problem.
2. Explain why the feedback in a bandit problem is evaluative rather than instructive. Why does the received reward not directly reveal the best action?
3. Explain the difference between the true action value $q_*(a)$ and the estimate $Q_t(a)$. What kind of experience is needed for $Q_t(a)$ to become reliable?
4. A bandit has three actions. The observed rewards are $a_1 : 2, 4, 3$; $a_2 : 1, 5$; and $a_3 : 0, 2, 2, 4$. Compute the sample-average estimate for each action and identify the greedy action.
5. In a 5-armed bandit, suppose $\epsilon = 0.20$ and there is one greedy action. What is the probability of random exploration? What is the total probability that the greedy action is selected?
6. In ϵ -greedy action selection with two actions and $\epsilon = 0.50$, what is the total probability that the greedy action is selected? Show the calculation.

7. A 4-armed bandit uses ϵ -greedy action selection and sample-average estimates with initial values $Q_1(a) = 0$ for all actions. The first observations are $A_1 = 1, R_1 = -1$; $A_2 = 2, R_2 = 1$; $A_3 = 2, R_3 = -2$; $A_4 = 2, R_4 = 2$; $A_5 = 3, R_5 = 0$. Identify the time steps on which random exploration definitely occurred, and the time steps on which it could have occurred.
8. A website can show one of four banners to a visitor. A click gives reward 1 and no click gives reward 0. Identify the arms and rewards in this bandit formulation. What is not observed after a banner is selected?
9. A cloud platform tests three cache configurations. The observed performance scores are $c_1 : 72, 76, 74$; $c_2 : 70, 80$; and $c_3 : 68, 69, 71$. Compute the current action-value estimate for each configuration and identify the greedy configuration.
10. Based on the 10-armed testbed curves, explain why the greedy method can improve quickly at first but still perform poorly in the long run.
11. Compare $\epsilon = 0.1$ and $\epsilon = 0.01$ in the 10-armed testbed. Which one explores more often, and how does that affect early discovery of the optimal action?

10. Points to Remember

- Bandit problems study repeated action selection in a single situation under evaluative feedback.
- The true value $q_*(a)$ is unknown; the learner estimates it through rewards observed after selecting actions.
- Sample-average estimation is a simple way to estimate action values from past rewards.
- Greedy selection is best according to current estimates, but it can stop testing alternatives too early.
- ϵ -greedy selection provides a simple mechanism for occasional exploration.
- The 10-armed testbed demonstrates the practical value of exploration when early rewards are noisy.

11. Required Reading

Sutton, R. S., and Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press. Required reading: **Chapter 2, Sections 2.1 to 2.3**, covering the k -armed bandit problem, action-value methods, greedy and ϵ -greedy action selection, and the 10-armed testbed.

12. References

Sutton, R. S., and Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). Cambridge, MA: MIT Press.

13. Acknowledgements

These lecture notes are based on and adapted from Chapter 2 of Sutton and Barto's *Reinforcement Learning: An Introduction*, Second Edition.

End of Topic -2 (Part 1)